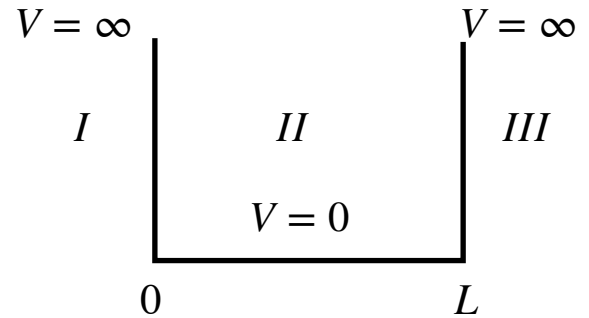


PROBLEM SET 5

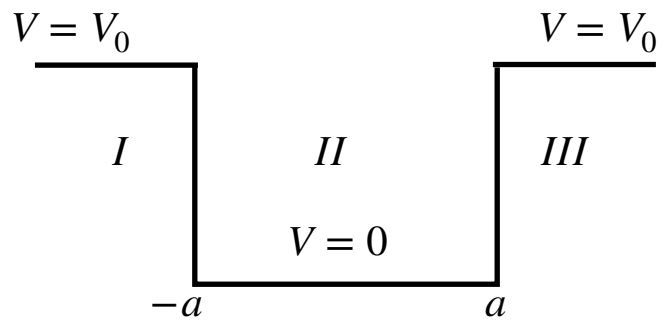
Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 30)
(**Extra Credit Max Score:** 15)

- 1) **(5 pts)** A particle is in the ground state of a box of length L . Suddenly the box expands (symmetrically) to twice its size, leaving the wave function undisturbed. Show that the probability of finding the particle in the ground state of the new box is $(8/3\pi)^2$.

- 2) **(10 pts)** In the **particle in a box** problem we solved in class, if the box extends from $x = 0$ to L (instead of $-L/2$ to $+L/2$) show that $\psi_n(x) = (2/L)^{1/2} \sin(n\pi x/L)$, $n = 1, 2, 3, \dots, \infty$ and $E_n = \hbar^2 \pi^2 n^2 / 2mL^2$.



3) **(15 pts)** Consider a particle in a square well potential:

$$V(x) = \begin{cases} 0 & |x| < a \\ V_0 & |x| \geq a \end{cases}$$


The diagram shows a square well potential. The potential is zero for $|x| < a$ (region II) and V_0 for $|x| \geq a$ (regions I and III). The well is bounded by $-a$ and a . The potential is labeled $V = V_0$ outside the well and $V = 0$ inside the well.

Since when $V_0 \rightarrow \infty$, we have a box, let us guess what the lowering of the walls does to the states. First if all, all the bound states (which alone we are interested in), will have $E \leq V_0$. Second, the wave function of the low-lying levels will look like those of the particle in a box, with the obvious difference that ψ will not vanish at the walls but instead spill out with an exponential tail. The eigenfunctions will still be even, odd, even, etc.

(a) **(5 pts)** Show that the even solutions have energies that satisfy the transcendental equation: $k \tan ka = \kappa$

(b) **(5 pts)** Show that the odd solutions will have energies that satisfy $k \cot ka = -\kappa$,

where k and $i\kappa$ are the real and complex wave numbers inside and outside the well, respectively. Note that k and κ are related by $k^2 + \kappa^2 = 2mV_0/\hbar^2$

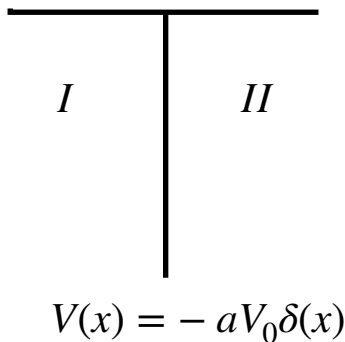
(c) **(5 pts)** Verify that as V_0 tends to ∞ , we regain the energy levels in the box

Extra Credit

(Points obtained here will be added to Mid-term)

- 4) Xtra Credit 1 **(10 pts)** Consider the potential $V(x) = -aV_0\delta(x)$, which is a Dirac-delta function.

Show that it admits a bound state of energy $E = -ma^2V_0^2/2\hbar^2$.



Hint: Solve Schrodinger's equation outside the potential for $E < 0$, and keep only the solution that has the right behavior at infinity and is continuous at $x = 0$. Calculate the change in slope evaluated in each of the two regions (i.e. $\psi'_{II}(\epsilon) - \psi'_{I}(-\epsilon)$), and equate it to $\int_{-\epsilon}^{\epsilon} \left(\frac{d^2\psi}{dx^2} \right) dx$

where ϵ is a small number nearby the delta function whose limit can be taken to $\epsilon \rightarrow 0$ when approaching the delta function.

5) Xtra Credit 2 **(5 pts)** Consider a particle of mass m in the state $|n\rangle$ of a box of length L .

- (a) **(1 pt)** Find the force $F = -\partial E/\partial L$ encountered when the walls are slowly pushed in, assuming the particle remains in the n th state of the box as its size changes.

Now consider a classical particle of energy $E_n = p^2/2m = (1/2)mv^2$ in this box.

Find a general expression for:

- (b) **(1 pt)** the change in momentum ΔP or *momentum transferred* for a particle colliding with the wall (think of momentum right before and after particle collides with wall, and also consider direction)
- (c) **(1 pt)** show the frequency of collision on a given wall (or equivalently, the average time interval $\Delta \bar{t}$ it takes for the particle to hit the same wall twice) is given by $\Delta \bar{t} = 2L/v$, where v is the speed of the particle.
- (d) **(2 pts)** calculate the average force, $\bar{F} = \Delta P/\Delta \bar{t}$, and show it is in agreement with the result obtained in the quantum case in part (a)